

Decomposing the Effect of Classification Fine-Tuning on Class Activation Maps via Head–Backbone Weight Swapping

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Abstract

Weakly supervised object localization (WSOL) methods build on class activation mapping (CAM) and are typically trained by fine-tuning an ImageNet-pre-trained classifier, yet whether this fine-tuning makes the CAM localize objects better is hard to measure directly, because a randomly initialized target head leaves the pre-trained network without a CAM for the target classes. We remove this obstacle with Stanford Dogs, whose 120 breeds are all ImageNet-1k classes: initializing the 120-way head from the corresponding ImageNet rows gives the pre-trained and fine-tuned ResNet-50 a shared class axis, so the CAM of the same class on the same image can be compared before and after fine-tuning, and exchanging heads and backbones between the two models decomposes the change without further training. On 7,985 single-object test images, full fine-tuning contracted the CAM and lowered MaxBoxAccV2 from 75.09% to 74.53%, but raised the energy ratio from 0.763 to 0.793, although most of the energy gain was not reproduced by an offset-invariant variant of the ratio. The backbone accounted for 0.99 and 1.00 of these two changes, respectively, and the fine-tuned head hardly changed the CAM. The direction was the same on ILSVRC2012 validation images of the 120 breeds, and in a supplementary setting that doubled the map resolution without retraining the MaxBoxAccV2 change reversed sign. Whether fine-tuning improved CAM localization thus depended on the metric, and we recommend reporting a threshold-free box metric, a mass-based metric, and a prior-map baseline together, along with the map resolution.

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1. Introduction

Class activation mapping (CAM) turns a classifier whose last feature map is globally average pooled and fed to a single linear layer into a class-specific score map: summing the feature maps with the linear weights of a class highlights the image regions that the network uses to identify that class (Zhou et al., 2016). Most weakly supervised object localization (WSOL) methods build on CAM (Wang et al., 2021), and they are typically trained by fine-tuning an ImageNet-pre-trained network for classification on the target dataset (Choe et al., 2020; Zhang et al., 2020). Rather than proposing a WSOL method, this paper asks how this fine-tuning step changes the CAM and whether the fine-tuned CAM localizes objects better than the CAM of the same network before fine-tuning.

Prior work does not settle the answer. The classification loss optimizes localization only indirectly (Zhang et al., 2020; Wang et al., 2021), and CAMs fail in two opposite ways: they concentrate on the most discriminative parts, such as the head of an animal, and underestimate the object (Zhou et al., 2016; Choe et al., 2023), or they activate on the background and overestimate it (Wang et al., 2021). Across WSOL methods, classification and localization performance are generally uncorrelated (Choe et al., 2023). Saliency maps have been observed to become more focused as fine-tuning raised classification accuracy from 22.0% to 90.1%, but only qualitatively (Wang et al., 2020).

A direct comparison is difficult to set up. When the target classes differ from the pre-training classes, fine-tuning replaces the classification head with a randomly initialized one, so the network before fine-tuning has no CAM for a target class (Section 3.2). We avoid this problem with Stanford Dogs, whose 120 breeds all correspond to ImageNet-1k classes. Initializing the 120-way head with the corresponding rows of the ImageNet head gives the pre-trained and fine-tuned models a shared class axis: the model at the start of fine-tuning is identical to the pre-trained model on these classes, and the CAM of the same class on the same image can be compared before and after fine-tuning (Figure 1). Section 3.1 states the four research questions.

Our contributions are as follows.

- A controlled before-and-after design built on this shared class axis, evaluated against no-learning baselines that include a prior map of box locations (Sections 3.2 and 3.4).
- A decomposition of the fine-tuning change into head, backbone, and interaction terms by exchanging heads and backbones between models, which the shared class axis allows without further training. The decomposition is exact at the score-map level (Equation (2)) and is evaluated at the metric level with an explicit interaction term (Section 3.3).
- Measurements with ResNet-50 on Stanford Dogs (Section 4). Full fine-tuning contracted the CAM and raised the energy ratio but lowered MaxBoxAccV2. The change came almost entirely from the backbone, and the fine-tuned head hardly changed the CAM. The direction was the same on ILSVRC2012 validation images of the 120 breeds.

2. Background and Related Work

WSOL methods and their evaluation. WSOL methods since CAM have tried to extend the map beyond the most discriminative parts through architectural changes or data augmentation. [Choe et al. \(2020\)](#) showed that much of their reported progress came from implicit full supervision in model selection and from a fixed score-map threshold, usually 0.2. Under a protocol with a small fully annotated validation set and threshold-free metrics, these methods did not improve significantly over CAM, and methods that enlarged the score maps did not necessarily yield boxes with higher IoU at the optimal threshold.

Training objectives and trained layers. [Zhang et al. \(2020\)](#) proposed to separate class-agnostic localization from classification; on all five backbones they tested, models trained jointly for both tasks performed worse than separately trained ones. They also reported that on CUB a VGG-GAP CAM reached 57.96% GT-known localization accuracy when all layers were fine-tuned and 57.53% when only the last linear layer was fine-tuned on top of ImageNet features. [Wang et al. \(2021\)](#) enlarged or shrank CAMs by updating only the linear layer after GAP with dedicated regularizers while the backbone was frozen. These studies compare training objectives or final models; to our knowledge, none of them relates a fine-tuned model to its own pre-trained starting point or separates the head and backbone contributions to the change caused by fine-tuning.

3. Method

3.1. Problem Setup

Let f be the convolutional backbone of a classifier whose last feature map is globally average pooled (GAP) and fed to a single linear layer with weights $\mathbf{W} \in \mathbb{R}^{C \times K}$ and bias $\mathbf{b} \in \mathbb{R}^C$. For an image x , $F = f(x) \in \mathbb{R}^{K \times H \times W}$ is the last feature map and the logit of class c is $z_c = \mathbf{w}_c^\top \text{GAP}(F) + b_c$, where $\mathbf{w}_c$ is the c -th row of $\mathbf{W}$. Following [Zhou et al. \(2016\)](#), the class activation map (CAM) of class c is

$$M^c(i, j) = \sum_{k=1}^K w_{c,k} F_k(i, j), \quad (1)$$

which we write as $M^c = \mathbf{w}_c^\top F$. The bias is omitted because it is spatially uniform. We use ResNet-50 ([He et al., 2016](#)), for which $K = 2048$ and $H = W = 7$ at an input size of 224×224 .

We evaluate GT-known localization ([Choe et al., 2020; 2023](#)): for an image with ground-truth class c^* and box annotation $\mathcal{B}$, the CAM M^{c^*} is compared with $\mathcal{B}$ whether or not the classifier predicts c^* , which separates localization from classification. We compare a network before fine-tuning (*Pre*) with the same network after classification fine-tuning on a target dataset and ask four questions. **RQ1:** Does fine-tuning improve or degrade CAM localization, and does the CAM expand or contract? **RQ2:** Does the change come from the classification head or from the backbone features? **RQ3:** Does localization change in the same direction as classification accuracy during training? **RQ4:** Do the answers depend on object size, on whether *Pre* classifies the image correctly, and on the evaluation set? RQ1 is confirmatory, RQ2 secondary, RQ3 descriptive, and RQ4 exploratory.

3.2. A Shared Class Axis Between Pre-Trained and Fine-Tuned Models

As noted in Section 1, a randomly initialized head leaves the network before fine-tuning without a CAM for a target class. Surrogates such as the CAM of the predicted source class or a class-agnostic map built from principal components of the features, as used by [Zhang et al. \(2020\)](#), are computed with weight vectors different from those of the fine-tuned CAM, so a difference between the two cannot be attributed to fine-tuning alone.

We avoid this problem by choosing a target dataset whose classes are a subset of the pre-training classes. Stanford Dogs ([Khosla et al., 2011](#)) was built from ImageNet leaf synsets under “Canis familiaris”, and ILSVRC2012 includes 120 dog breeds ([Russakovsky et al., 2015](#)). Matching the WordNet IDs of the two class lists, all 120 Stanford Dogs breeds correspond to ImageNet-1k classes. We therefore

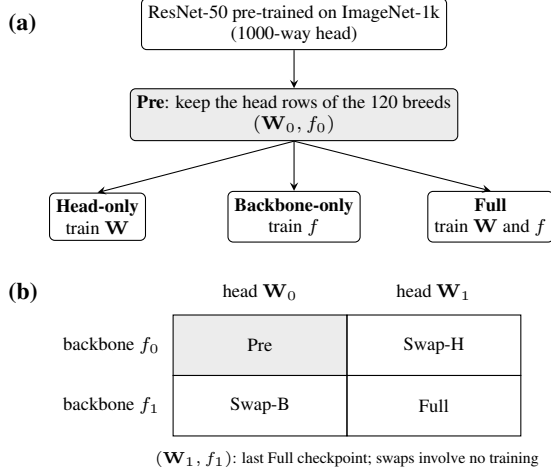


Figure 1. Experimental design. (a) Every model starts from Pre, an ImageNet-pre-trained ResNet-50 whose head keeps the rows of the 120 Stanford Dogs breeds, and the conditions differ only in which parameters are fine-tuned. (b) Weight swaps recombine the head $\mathbf{W}$ and the backbone f (including its batch-normalization statistics) of Pre and of the last Full checkpoint.

replace the 1000-way head of an ImageNet-pre-trained ResNet-50 with a 120-way head whose weights and biases are copied from the 120 corresponding rows. This construction has three consequences (Figure 1):

1. The model at epoch 0 of fine-tuning is exactly Pre restricted to the 120 breeds. Its logits matched the corresponding entries of the original 1000-way logits (absolute tolerance 10^{-4} on 16 images).
2. Pre and every fine-tuned model yield a CAM for the same class on the same image, so the two can be compared directly.
3. All models share one class axis, so heads and backbones from different models can be recombined without further training (Section 3.3).

3.3. Fine-Tuning Conditions and Weight Swapping

Training conditions. All conditions start from Pre and differ only in the trained parameters (Figure 1a). *Head-only* trains $\mathbf{W}$ and $\mathbf{b}$. Its backbone is kept in evaluation mode, which freezes the batch-normalization (BN) running statistics as well, so its features are identical to those of Pre. *Backbone-only* trains all convolution and BN parameters and updates the BN statistics, while the head stays at its ImageNet initialization. *Full* trains all parameters, as in standard fine-tuning.

Weight swapping. The training conditions reach different end points after different amounts of parameter change, so

the Head-only and Backbone-only changes need not add up to the Full change. To decompose the Full change along a single training trajectory, we recombine the weights of Pre, $(\mathbf{W}_0, f_0)$, and of the last Full checkpoint, $(\mathbf{W}_1, f_1)$ (Figure 1b). *Swap-H* uses $(\mathbf{W}_1, f_0)$ and *Swap-B* uses $(\mathbf{W}_0, f_1)$, where a backbone includes its BN statistics. Because Equation (1) is bilinear in the head weights and the features, the change from Pre to Full decomposes exactly at the level of the score map. Writing $F_0 = f_0(x)$ and $F_1 = f_1(x)$ and dropping the class index,

$$\begin{aligned} & \mathbf{w}_1^\top F_1 - \mathbf{w}_0^\top F_0 \\ &= \underbrace{(\mathbf{w}_1 - \mathbf{w}_0)^\top F_0}_{\text{Swap-H} - \text{Pre}} + \underbrace{\mathbf{w}_0^\top (F_1 - F_0)}_{\text{Swap-B} - \text{Pre}} \\ &+ \underbrace{(\mathbf{w}_1 - \mathbf{w}_0)^\top (F_1 - F_0)}_{\text{interaction}}. \end{aligned} \quad (2)$$

Localization metrics apply min-max normalization, thresholding, or a ReLU to the map, so this identity does not carry over to a metric m . With $\Delta_m(\cdot) = m(\cdot) - m(\text{Pre})$, we therefore compute the metric-level interaction

$$I_m = \Delta_m(\text{Full}) - \Delta_m(\text{Swap-H}) - \Delta_m(\text{Swap-B}). \quad (3)$$

We read $\Delta_m(\text{Swap-H})/\Delta_m(\text{Full})$ and $\Delta_m(\text{Swap-B})/\Delta_m(\text{Full})$ as head and backbone contributions only if the 95% confidence interval (CI) of $\Delta_m(\text{Full})$ excludes zero and the 95% CI of I_m lies within $\pm 0.25 |\Delta_m(\text{Full})|$. If only the first condition holds, we report the interaction itself; if the CI of $\Delta_m(\text{Full})$ contains zero, we report the three changes without ratios. The tolerance of 0.25 is arbitrary. Swaps are the primary evidence for RQ2, and Head-only and Backbone-only serve as a check on the direction of the effects.

3.4. Evaluation Metrics

Processing. The 7×7 map M^{c*} is bilinearly upsampled to 224×224 , giving $\tilde{M}$. Threshold-based metrics use the per-image min-max normalized map $\hat{M} = (\tilde{M} - \min \tilde{M}) / (\max \tilde{M} - \min \tilde{M})$, following Choe et al. (2020), and mass-based metrics use $\tilde{M}^+ = \max(\tilde{M}, 0)$. Upsampling and the ReLU do not commute, so this order is fixed for every model. Let Ω be the set of all pixels and $\Omega_B \subseteq \Omega$ the set of pixels inside the ground-truth box.

Co-primary metrics. *MaxBoxAccV2* (Choe et al., 2023) measures box localization without committing to a threshold. Let $h_n(\tau, \delta) = \mathbf{1}[\text{IoU}(\text{boxes}(\hat{M}^{(n)}, \tau), \mathcal{B}^{(n)}) \geq \delta]$ for image n , where $\text{boxes}(\hat{M}, \tau)$ is the set of tightest boxes around the 4-connected components of $\{\hat{M} \geq \tau\}$ and the IoU between two sets of boxes is the maximum over all

pairs. Then

$$\text{MaxBoxAccV2} = \frac{1}{3} \sum_{\delta \in \{0.3, 0.5, 0.7\}} \max_{\tau} \frac{1}{N} \sum_{n=1}^N h_n(\tau, \delta), \quad (4)$$

with $\tau \in \{0, 0.01, \dots, 0.99\}$ and the maximum taken separately for each δ . The *energy ratio* of the energy-based pointing game (Wang et al., 2020) is

$$E = \sum_{p \in \Omega_B} \tilde{M}_p^+ / \sum_{p \in \Omega} \tilde{M}_p^+. \quad (5)$$

It needs no threshold, lies in $[0, 1]$, and is invariant to positive rescaling of the map. For a uniform map it equals the fraction of the image covered by the box. Wang et al. (2020) excluded objects larger than half of the image; we keep them because the boxes in our data cover 55.8% of the image on average (uniform-map baseline in Table 1), and we analyze object size explicitly (Section 4.4). MaxBoxAccV2 enables comparison with the WSOL literature, while E reflects the whole activation distribution.

Diagnostics. *Offset.* E is invariant to rescaling but, because of the ReLU, sensitive to an additive offset, whereas MaxBoxAccV2 is invariant to both. We therefore also compute E_{med} , which replaces $\tilde{M}^+$ in Equation (5) with $\max(\tilde{M} - \text{med}(\tilde{M}), 0)$ and is invariant to both. If the two co-primary metrics change in opposite directions, the CI of ΔE excludes zero, and the CI of ΔE_{med} contains zero, the disagreement is classified as an offset effect. *Expansion or contraction.* For $S_{\tau} = \{p : \hat{M}_p \geq \tau\}$ we compute the area $|S_{\tau}|/|\Omega|$, the recall $|S_{\tau} \cap \Omega_B|/|\Omega_B|$, and the precision $|S_{\tau} \cap \Omega_B|/|S_{\tau}|$, and average each over $\tau \in \{0.05, 0.06, \dots, 0.95\}$ for every image (area, recall, and precision AUC). The endpoints are excluded because min-max normalization makes all models identical at $\tau = 0$ and $\tau = 1$. A change is classified as a contraction if the CIs of Δ for both area AUC and recall AUC lie below zero, as an expansion if both lie above zero, and as undetermined otherwise; precision AUC then indicates whether the change moves activation toward the object or toward the background. *Concentration and relocation.* A_{50} is the smallest fraction of pixels, taken in decreasing order of $\tilde{M}^+$, that holds half of the total positive mass; a smaller A_{50} means a more concentrated map. Relocation is measured by the per-image Pearson correlation r between the 7×7 raw CAMs of Pre and another model, which ignores scale and offset.

No-learning baselines. Scores on datasets whose objects are large and centered can be high without any learning: on CUB, CAM trained with randomly chosen hyperparameters often scores below a center-Gaussian baseline (Choe et al., 2023), and chance localization on ILSVRC correlates

strongly with object size (Russakovsky et al., 2015). We therefore report two box baselines, the full-image box and a box of the mean size of the single-object training boxes placed at the image center, and two map baselines, a uniform map and a *prior map*. The prior map is the mean of the binary box masks of the 10,800 training images (union of boxes for multi-box images); the same map is used for every test image and captures both the size and the center bias. Maps are scored with the same pipeline as CAMs. Fixed boxes need no threshold, so their accuracy at each δ is reported directly and averaged in place of MaxBoxAccV2.

Relation to Grad-CAM. For a GAP-linear head, Grad-CAM (Selvaraju et al., 2017) equals $\text{ReLU}(M^c)/(HW)$, a positive multiple of the ReLU-CAM (Wang et al., 2020), so we do not evaluate it separately. Our threshold-based metrics use the min-max normalized raw CAM, so their values are not directly comparable with Grad-CAM-based results.

3.5. Experimental Protocol

Data. Stanford Dogs contains 120 breeds with 100 training images per class and 8,580 test images, all annotated with boxes (Khosla et al., 2011). We hold out 10 training images per class (1,200 images) as a validation split, the same per-class budget of fully annotated images that Choe et al. (2020) allow for ImageNet, and train on the remaining 10,800. All choices are made on the validation split, and the test split is used only for final reporting. Main analyses use the 7,985 single-object test images; the 595 multi-object images are analyzed separately (Section 4.4). As a second evaluation set we use the ILSVRC2012 validation images of the same 120 classes (50 per class), in which every instance of the ground-truth class is annotated (Russakovsky et al., 2015). We remove 990 multi-object images and 3 images identical to Stanford Dogs training images, leaving 5,007 images. We did not check whether Stanford Dogs test images overlap with the ImageNet-1k training images used for pre-training, and the gap in Pre top-1 accuracy between the two evaluation sets (Section 4.3) is not explained by our analysis. Images are resized to 224×224 without cropping, so that no box is truncated, and boxes are rescaled accordingly. Pre is evaluated with the same preprocessing; under the torchvision default (resize to 256, center crop to 224) its Stanford Dogs test accuracy would be 91.35% instead of 88.74%.

Training. All conditions minimize the cross-entropy loss with SGD (momentum 0.9, weight decay 10^{-4} , batch size 32) for 30 epochs, dividing the learning rate by 10 at epochs 10 and 20. Prior WSOL fine-tuning applies larger learning rates to higher layers (Choe et al., 2020) or to newly added, randomly initialized components (Zhang et al.,

2020); we apply no layer-wise multiplier because our head starts from ImageNet weights. Augmentation consists of horizontal flipping and an isotropic rescaling about the image center with a factor drawn from $U(0.9, 1.1)$; the rescaling was added after a pilot Full run without it showed a 13.8 pp gap between training and validation top-1 accuracy. The learning rate of each condition was selected by final-epoch validation top-1 accuracy (seed 0) over log-spaced grids, $\{3 \times 10^{-4}, \dots, 3 \times 10^{-2}\}$ for Head-only and $\{3 \times 10^{-5}, \dots, 3 \times 10^{-3}\}$ for the others. Each grid initially had four values and was extended once downward after the lowest value was selected for every condition; the added values were not selected. The selected rates are 10^{-3} for Head-only and 10^{-4} for Backbone-only and Full. In 1,000 class-stratified bootstrap resamples of the validation split, the same rates were selected 74.4%, 82.3%, and 96.2% of the time, respectively, so the Head-only selection was less stable. Each condition is then trained with seeds 1, 2, and 3; because all runs share the same initialization, the seeds change only the data order and augmentation, apart from nondeterminism of cuDNN kernels. The seed-0 runs are not used in the analysis. We evaluate the last checkpoint, as recommended by Choe et al. (2023). All experiments ran on a single NVIDIA GeForce RTX 5070 Ti Laptop GPU with PyTorch 2.11.0 and torchvision 0.26.0 (CUDA 12.8).

Statistical analysis. For a metric m and a condition trained with seeds $s \in \{1, 2, 3\}$, the estimate is $\bar{\Delta} = \frac{1}{3} \sum_s (m(\text{cond}_s) - m(\text{Pre}))$, computed on the same images. Uncertainty is estimated with 10,000 image-level bootstrap resamples, a procedure also used to compare ILSVRC entries (Russakovsky et al., 2015). Within an evaluation set, all conditions, seeds, and metrics share the same resamples, so $\Delta(\text{Full})$, $\Delta(\text{Swap-H})$, $\Delta(\text{Swap-B})$, and I_m are resampled jointly; for MaxBoxAccV2 the maximum over τ is recomputed in every resample. We report percentile CIs and two-sided bootstrap p -values, whose smallest attainable value is $2/10,001 \approx 2 \times 10^{-4}$. The confirmatory hypotheses for RQ1 are that Full differs from Pre in MaxBoxAccV2 and in E on the single-object Stanford Dogs test images, tested with the Holm–Bonferroni procedure at a family-wise level of 0.05; the CIs are reported at the matching levels of 97.5% and 95%. A conclusion additionally requires that the per-seed changes of all three seeds share the sign of $\bar{\Delta}$, because the bootstrap captures only image sampling. With three seeds this is weak evidence about training randomness, since a random sign would agree across all three seeds with probability $1/4$. For RQ2 the Holm procedure is applied to the six tests $\{\text{Swap-H}, \text{Swap-B}, I_m\} \times \{\text{MaxBoxAccV2}, E\}$; RQ3 involves no tests, and RQ4 reports unadjusted 95% CIs. From the validation split we estimated minimum detectable effects on the test set of 0.42 percentage points (pp) for MaxBoxAccV2 and 0.00075 for E (two-sided $\alpha = 0.025$, power 0.8, standard error scaled by $\sqrt{1123/7985}$).

4. Results

Unless stated otherwise, results refer to the 7,985 single-object Stanford Dogs test images, values are means over three seeds, and $\bar{\Delta}$ is a model minus Pre. MaxBoxAccV2 and accuracies are reported in percent and their differences in pp. Table 1 summarizes all models and baselines. Our evaluation set, CAM resolution, and implementation details (connected-component rule and threshold grid) differ from WSOL benchmarks, so absolute MaxBoxAccV2 values should not be compared with published benchmark numbers; every comparison below is made against the same Pre model and the same baselines.

4.1. Fine-Tuning Concentrates the CAM but Lowers Box Accuracy (RQ1)

Confirmatory tests. Both confirmatory tests were significant, with effects in opposite directions (Table 2). Full fine-tuning lowered MaxBoxAccV2 from 75.09% to 74.53% ($\bar{\Delta} = -0.56$ pp, 97.5% CI $[-0.87, -0.32]$) and raised the energy ratio from 0.763 to 0.793 ($\bar{\Delta} = +0.0300$, 95% CI $[+0.0295, +0.0305]$). For both metrics $p = 2 \times 10^{-4}$, the smallest attainable value, and all three seeds changed in the same direction, so both conclusions satisfy the decision rule of Section 3.5. Both changes exceed the minimum detectable effects given there, the MaxBoxAccV2 change only modestly and the energy change by a factor of about 40.

The disagreement is not an offset artifact. Because the two co-primary metrics moved in opposite directions, we applied the offset diagnostic. The offset-invariant E_{med} also increased ($+0.0016$, 95% CI $[+0.0012, +0.0020]$), so the disagreement is not classified as an offset effect. The increase in E_{med} is, however, only about 5% of that in E , and relative to the prior map the margin grew by 33% for E but by 7.5% for E_{med} (Table 1). Most of the gain in E is thus not reproduced by the offset-invariant E_{med} and may reflect a change in the negative part of the map, which the ReLU discards.

The CAM contracts. Area AUC and recall AUC decreased and precision AUC increased, all with CIs excluding zero (Table 2). By the rule of Section 3.4 the CAM *contracted*: at a given threshold the retained region was smaller, covered less of the box, and lay more inside the box. A_{50} fell from 0.232 to 0.204, a reduction of 12.3%. On the 7×7 feature map, where one cell covers $1/49 \approx 2.0\%$ of the image, this reduction corresponds to about 1.4 cells of the 7×7 map. The spatial layout was largely preserved, with a mean per-image correlation between Pre and Full CAMs of $r = 0.979$ (median 0.985). Fine-tuning therefore tightened the CAM around its existing peak rather than moving it.

Table 1. Classification and localization on Stanford Dogs test. Top-1 accuracy is 120-way (Pre is restricted to the 120 breeds) and computed on all 8,580 test images; localization uses the 7,985 single-object images. Model rows are means over three seeds. MaxBoxAccV2 and its per- δ components are in %. The energy ratio of the uniform map equals the mean box-area fraction.

Model	Top-1 (%)	Box accuracy (%)				Activation mass		
		MaxBoxAccV2	$\delta = 0.3$	$\delta = 0.5$	$\delta = 0.7$	E	E_{med}	A_{50}
<i>No-learning baselines</i>								
Full-image box	–	58.15	81.94	59.49	33.02	–	–	–
Mean-size center box	–	63.38	91.92	76.24	21.97	–	–	–
Uniform map	–	–	–	–	–	0.5582	–	–
Prior map	–	72.62	94.20	77.76	45.90	0.6730	0.8399	0.3305
Pre	88.74	75.09	97.48	82.82	44.97	0.7627	0.8611	0.2321
<i>Fine-tuned</i>								
Head-only	88.39	74.98	97.46	82.75	44.74	0.7628	0.8611	0.2318
Backbone-only	88.29	74.49	97.43	82.24	43.81	0.7922	0.8625	0.2041
Full	88.34	74.53	97.44	82.25	43.88	0.7927	0.8627	0.2037
<i>Weight swaps (no training)</i>								
Swap-H	88.22	75.08	97.47	82.83	44.93	0.7629	0.8611	0.2319
Swap-B	88.10	74.53	97.44	82.28	43.89	0.7926	0.8627	0.2038

Table 2. Full – Pre on Stanford Dogs test (single-object). Upper block: the two confirmatory tests, with CI levels matching the Holm-adjusted significance levels ($\dagger 97.5\%$, otherwise 95%). Lower block: diagnostics with 95% CIs, including the offset-invariant E_{med} . Every p equals 2×10^{-4} , the smallest attainable with 10,000 resamples, and all three seeds share the sign of $\bar{\Delta}$.

Metric	$\bar{\Delta}$	CI
MaxBoxAccV2 (pp) †	–0.565	[–0.870, –0.319]
Energy ratio E	+0.0300	[+0.0295, +0.0305]
E_{med}	+0.0016	[+0.0012, +0.0020]
Area AUC	–0.0159	[–0.0167, –0.0152]
Recall AUC	–0.0190	[–0.0199, –0.0181]
Precision AUC	+0.0077	[+0.0072, +0.0081]
A_{50}	–0.0284	[–0.0288, –0.0281]

The box-accuracy loss grows with the IoU threshold. At $\delta = 0.3, 0.5$, and 0.7 the Full – Pre change was -0.04 , -0.56 , and -1.09 pp (Table 1). MaxBoxAccV2 selects, for each model and δ , one threshold shared by all images, so a contraction that acted as the same monotone transformation on every map would be offset by a lower threshold. The loss at strict IoU is not recovered by re-selecting a single global threshold; our measurements do not identify which change causes it.

The margin over dataset bias is small. Every model exceeded the prior map on both co-primary metrics. For MaxBoxAccV2, however, the margin shrank from 2.47 pp for Pre to 1.91 pp for Full, and at $\delta = 0.7$ the prior map (45.90%) exceeded even Pre (44.97%). For the energy ratio the margin grew from 0.090 to 0.120. The fixed-box baselines were far lower (58.15% and 63.38%).

Qualitative examples. Figure 2 shows six test images drawn at random. The Pre and Full maps are hard to tell apart by eye, consistent with the mean correlation of 0.979. Their differences are at most 0.266 on the normalized scale and vary smoothly over the image, without a single pattern across the six images: in the first three columns activation increased near the Pre peak and decreased farther from it, whereas the fourth column changed little and the sixth gained activation at the bottom of the box. Six images illustrate rather than establish the quantitative result.

4.2. The Change Comes from the Backbone (RQ2)

Figure 3 decomposes the Pre-to-Full change with the weight swaps; CIs in this subsection are 95% and not adjusted to the Holm levels. Replacing only the backbone (Swap-B) reproduced almost the entire change: -0.557 pp in MaxBoxAccV2 (95% CI [–0.822, –0.341]) and $+0.0299$ in the energy ratio ([+0.0294, +0.0304]), against -0.565 pp and $+0.0300$ for Full. Replacing only the head (Swap-H) changed MaxBoxAccV2 by -0.013 pp ([–0.070, +0.032], $p = 0.47$) and the energy ratio by $+0.00024$. The interaction was $I_m = +0.004$ pp ([–0.057, +0.064]) for MaxBoxAccV2 and -0.00014 ([–0.00015, –0.00012]) for the energy ratio, inside the tolerances of ± 0.141 pp and ± 0.0075 . Both metrics therefore meet the rule for reading the swaps as contributions: the backbone accounts for 0.99 (MaxBoxAccV2) and 1.00 (energy ratio) of the change and the head for 0.02 and 0.01. After Holm correction over the six RQ2 tests, Swap-B was significant for both metrics, whereas Swap-H and I_m were significant only for the energy ratio, where they amount to 0.8% and 0.5% of the Full change.

The training conditions agree in direction (point estimates at the last checkpoint, Table 1). Backbone-only, which never



Figure 2. CAMs of six Stanford Dogs test images drawn at random (single-object, random seed 0, no selection) for Pre and Full (seed 1). Each map is upsampled, min-max normalized per image ($\hat{M}$), and overlaid on the input; boxes are the ground truth. The bottom row shows $\hat{M}_{\text{Full}} - \hat{M}_{\text{Pre}}$ on a diverging colormap centered at zero, with limits ± 0.266 set by the largest absolute difference over the six images and shared by all columns. The weight swaps are not shown: the per-image mean absolute difference of $\hat{M}$ is at most 0.0032 between Swap-H and Pre and at most 0.0036 between Swap-B and Full, against 0.020–0.076 between Pre and Full.

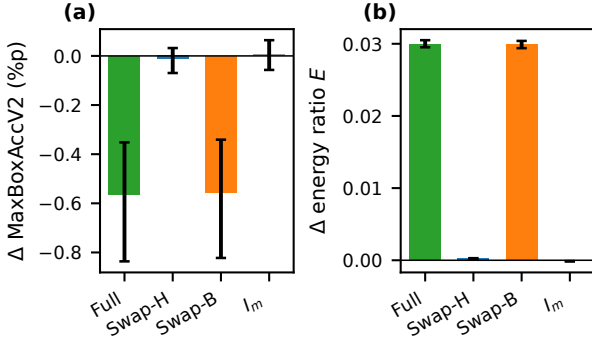


Figure 3. Decomposition of the Pre-to-Full change on Stanford Dogs test (single-object). Bars show Δ for Full, Swap-H (head replaced), Swap-B (backbone replaced), and the interaction I_m of Equation (3); error bars are 95% bootstrap CIs computed with shared resamples. (a) MaxBoxAccV2 in pp. (b) Energy ratio.

updated the head, changed MaxBoxAccV2 by -0.60 pp and the energy ratio by $+0.0295$, close to Full. Head-only changed them by only -0.11 pp and $+0.00014$, even though its learning rate was ten times that of Full.

Changing the head left the spatial pattern of the CAM essentially intact. The mean per-image correlation with Pre was 0.9996 for Head-only and 0.99999 for Swap-H, against

0.979 for every model with a fine-tuned backbone. Swap-H had a lower test top-1 accuracy than Pre (88.22% vs. 88.74%); this is a point estimate without a CI, and part of it may come from the bias $\mathbf{b}$, which affects the logits but is excluded from the CAM (Section 3.1). Unlike re-weighting the linear layer with dedicated regularizers (cf. Section 2), plain cross-entropy fine-tuning from the ImageNet head barely altered the CAM.

4.3. Training Dynamics and Classification Accuracy (RQ3)

Setting. During training we recorded predictions and ground-truth-class CAMs on the validation split after every epoch, including epoch 0, and computed localization metrics at epochs 0, 5, $\dots$, 30 on its 1,123 single-object images; top-1 accuracy uses all 1,200 images. Observations along one trajectory are autocorrelated, so we report trajectories and directions without significance tests. The validation minimum detectable effects are 1.11 pp for MaxBoxAccV2 and 0.0020 for the energy ratio.

Trajectories. In all six runs that trained the backbone (Figure 4), the energy ratio was highest at epoch 5 among the evaluated epochs (0.796–0.798, from 0.759 at epoch 0), fell back by epoch 10, and stayed between 0.787 and 0.791

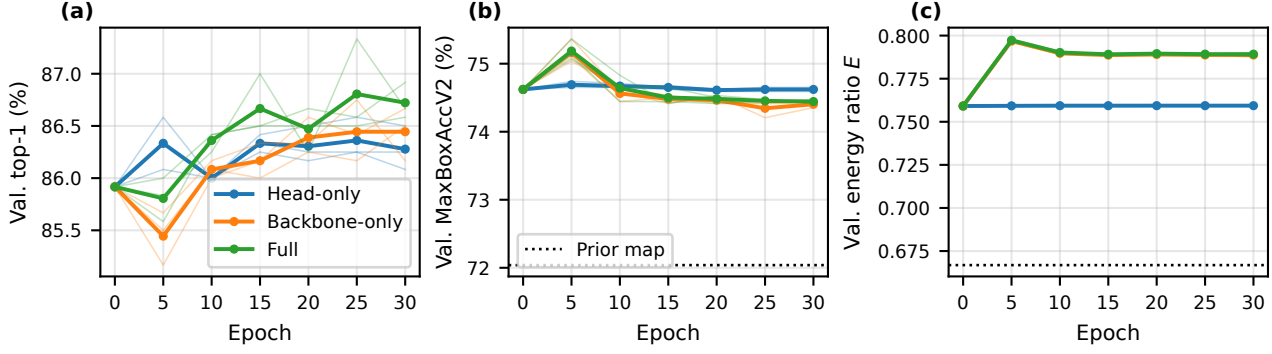


Figure 4. Trajectories on the validation split at epochs 0, 5, . . . , 30; epoch 0 is Pre for every condition. Thick lines are means over three seeds, thin lines individual seeds, and dotted lines the prior-map baseline. (a) Top-1 accuracy on 1,200 images. (b) MaxBoxAccV2 and (c) energy ratio on 1,123 single-object images. The movements in (b) are smaller than the validation minimum detectable effect (Section 4.3).

thereafter, well above Pre. MaxBoxAccV2 was also highest at epoch 5 in all six runs (+0.42 to +0.74 pp) and stayed below Pre from epoch 20 onward, but these movements are within the validation minimum detectable effect and we do not interpret them. Validation top-1 was within 1 pp of Pre at epoch 5, exceeded Pre by epoch 10, and ended 0.25 to 1.00 pp above Pre at epoch 30. Head-only left both localization metrics essentially unchanged throughout. From epoch 0 to 30, validation top-1 and the energy ratio moved in the same direction in all nine runs, whereas validation top-1 and MaxBoxAccV2 moved in opposite directions in all six backbone-trained runs; we treat the latter only as a description.

Test accuracy did not increase. The validation gain (from 85.92% for Pre to 86.72% for Full, mean over seeds) did not carry over to held-out evaluation. Test top-1 accuracy was lower for Full than for Pre, 88.34% vs. 88.74% on Stanford Dogs and 76.64% vs. 78.26% on the 5,997 ILSVRC2012 validation images of the 120 breeds (all except the 3 duplicates); these are point estimates, as we did not compute CIs for accuracy. Among the single-object Stanford Dogs test images, 271 to 291 images per seed changed from correct to incorrect and 240 to 245 from incorrect to correct. The validation gain may also be optimistic, because the learning rates were selected on the same split. We therefore do not describe our results as a trade-off between classification and localization. Choe et al. (2023) reported that localization performance fluctuates substantially in early epochs; in our setting the early peak in the energy ratio recurred in all six backbone-trained runs and exceeded the validation minimum detectable effect.

4.4. Robustness Across Subsets, Evaluation Sets, and Resolution (RQ4)

Subgroups. We split the test images by object size, using tertiles of the box-area fraction on the single-object training

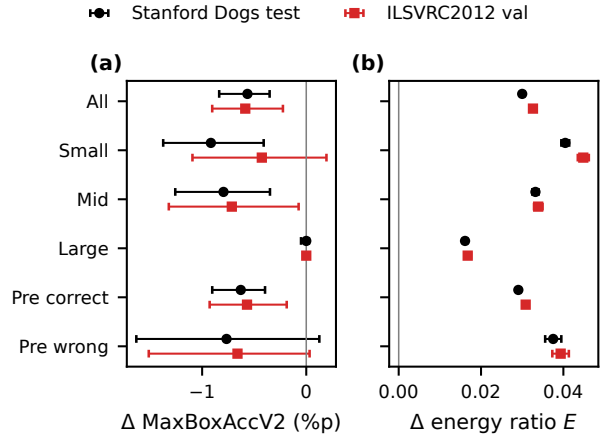


Figure 5. Full - Pre by subgroup, with unadjusted 95% bootstrap CIs. Size bins use tertiles of the box-area fraction on the training split. The Pre correct and Pre wrong groups are defined by the prediction of Pre and contain the same images for every model. (a) MaxBoxAccV2 in pp, with τ re-optimized within each subgroup; the large-object bin is exactly zero (Section 4.4). (b) Energy ratio.

images (0.449 and 0.699) so that the bin edges do not depend on test data, and by whether Pre classifies the image correctly. The latter groups are fixed by Pre and therefore contain the same images for every model. These analyses are exploratory and unadjusted. The energy gain decreased with object size on both evaluation sets (Figure 5b): +0.040, +0.033, and +0.016 for small, medium, and large objects on Stanford Dogs, and +0.045, +0.034, and +0.017 on the ILSVRC2012 subset, with non-overlapping CIs. A plausible reason is that large boxes leave little room between the uniform-map level and the upper bound of 1. The gain was also larger for images that Pre misclassified than for those it classified correctly (+0.038 vs. +0.029 on Stanford Dogs; +0.039 vs. +0.031 on ILSVRC2012); we did not examine how much this overlaps with the size effect.

Table 3. Robustness of the Full – Pre change (means over seeds, 95% bootstrap CIs). MaxBoxAccV2 is in % and its differences in pp. Pre and Full levels were not computed for multi-object images. Stride 1 removes the last downsampling of ResNet-50 without retraining.

Metric	Pre	Full	$\bar{\Delta}$ [95% CI]
<i>ILSVRC2012 val, 120 breeds, single-object (N = 5,007)</i>			
MaxBoxAccV2	74.46	73.87	−0.59 [−0.90, −0.22]
E	0.7632	0.7958	+0.0326 [+0.0319, +0.0333]
<i>Stanford Dogs test, multi-object (N = 595)</i>			
MaxBoxAccV2	–	–	−1.49 [−2.80, −0.26]
<i>Stanford Dogs test, single-object, stride 1 (14 × 14)</i>			
MaxBoxAccV2	76.20	76.80	+0.60 [+0.25, +1.02]
E	0.7981	0.8564	+0.0583 [+0.0574, +0.0592]

Within each subgroup, MaxBoxAccV2 re-optimizes the threshold τ on that subgroup alone. It decreased for small and medium objects on Stanford Dogs (−0.92 and −0.79 pp) and for medium objects on the ILSVRC2012 subset (−0.72 pp), whereas the CI for small objects on ILSVRC2012 and those for the Pre-wrong groups include zero (Figure 5a). In the large-object bin the change was exactly zero for every seed on both evaluation sets. This does not mean that the CAMs of large objects were unchanged. When the threshold is re-optimized within this bin, the best box for every model is the whole image, so bin-wise MaxBoxAccV2 cannot register a change for these objects: at $\tau = 0$ the accuracy at $\delta = 0.3, 0.5$, and 0.7 is 100%, 100%, and 99.36% on Stanford Dogs and 100%, 100%, and 99.46% on the ILSVRC2012 subset for Pre and for every Full seed.

Evaluation set. On the ILSVRC2012 subset, whose images come from the ImageNet validation split, the change had the same direction and a similar size (Table 3): −0.59 pp in MaxBoxAccV2 ($p = 8 \times 10^{-4}$) and +0.0326 in the energy ratio ($p = 2 \times 10^{-4}$). We do not attribute the remaining differences between the two sets to pre-training leakage, because the sets were also curated differently: Stanford Dogs removed degenerate images by hand (Khosla et al., 2011), and multi-object images make up 16.5% of the 6,000 ILSVRC2012 validation images of the 120 breeds against 6.9% of the 8,580 Stanford Dogs test images. On the 595 multi-object Stanford Dogs test images, where MaxBoxAccV2 matches all estimated boxes against all ground-truth boxes, MaxBoxAccV2 also decreased (−1.49 pp, $p = 0.02$).

Resolution. Because the contraction is about 1.4 cells of the 7×7 map, we repeated the analysis at 14×14 , one of the score-map resolutions used in WSOL (Choe et al., 2020), by setting the stride of the last ResNet-50 stage to 1 without retraining, for both Pre and Full. At 14×14 the

MaxBoxAccV2 change reversed sign (+0.60 pp, 95% CI [+0.25, +1.02], $p = 2.2 \times 10^{-3}$, same sign in all seeds), and the energy gain was larger than at 7×7 (+0.058 vs. +0.030; Table 3). The modification is not neutral: it lowered test top-1 accuracy from 88.74% to 83.83% for Pre and to 85.48–85.59% for Full, so the two models are perturbed by different amounts. We therefore treat the 14×14 result as supplementary; it suggests that the MaxBoxAccV2 conclusion of Section 4.1 may be specific to the native 7×7 resolution.

5. Discussion and Conclusion

Implications for evaluation. The energy ratio, which uses the whole positive mass of the map, rose, although most of the gain was not reproduced by the offset-invariant E_{med} ; MaxBoxAccV2, which scores boxes drawn from thresholded maps, fell at 7×7 . Choe et al. (2020) showed that a fixed score-map threshold can create illusory progress in WSOL and proposed threshold-free metrics to remove it; our results show that threshold-free metrics can still disagree in sign on the same change. Zhou et al. (2016) found that localization improved with a higher mapping resolution, and our supplementary, untrained stride-1 setting suggests that the resolution may change not only the level of MaxBoxAccV2 for Pre (75.09% at 7×7 , 76.20% at 14×14) but also the sign of the fine-tuning effect. At 7×7 the box metric also exceeded the prior map by only a few points (Section 4.1). We therefore recommend that studies of how training changes a CAM report a threshold-free box metric, a mass-based metric such as the energy ratio, and a prior-map baseline together, state the resolution of the map, and not infer localization quality from classification accuracy.

Scope of the conclusions. The conclusions rest on one backbone, ResNet-50, and on one family of data, the same 120 dog breeds in Stanford Dogs and ILSVRC2012. The design requires target classes that belong to the pre-training label set, which restricts the datasets it can use. Our fine-tuning also differs from common WSOL practice: the head started from ImageNet weights, the selected learning rate was 10^{-4} without layer-wise multipliers, and test top-1 accuracy did not increase (Section 4.3), so our results do not cover fine-tuning that improves classification. Future work can test whether the conclusions hold with layer-wise learning rates and for VGG and InceptionV3, the other two backbones used by Choe et al. (2020). The 14×14 maps were obtained by changing a stride without retraining, so models fine-tuned at a higher resolution are needed to tell whether the sign reversal reflects resolution itself.

Strength of the evidence.

- Three seeds give only weak evidence about training

randomness, and the bootstrap reflects image sampling only (Section 3.5); more seeds are needed.

- The interaction tolerance and the AUC range were not derived from data, and we did not analyze the sensitivity of the conclusions to them.
- We describe the change as a contraction with higher precision; whether the retained region is a breed-discriminative part requires part annotations.

Conclusion. By using a target dataset whose classes all belong to the pre-training label set, we compared the CAM of a network before and after classification fine-tuning for the same class on the same image, and decomposed the change into head and backbone contributions. Fine-tuning contracted the CAM and raised the energy ratio, most of which was not reproduced by the offset-invariant E_{med} ; the change came from the backbone features, and the fine-tuned head hardly changed the CAM. Whether this counts as better localization depended on the metric: it did for the energy ratio, and it did not for MaxBoxAccV2 at the native 7×7 resolution.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

References

- Choe, J., Oh, S. J., Lee, S., Chun, S., Akata, Z., and Shim, H. Evaluating weakly supervised object localization methods right. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3133–3142, 2020.
- Choe, J., Oh, S. J., Chun, S., Lee, S., Akata, Z., and Shim, H. Evaluation for weakly supervised object localization: Protocol, metrics, and datasets. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(2):1732–1748, 2023. doi: 10.1109/TPAMI.2022.3169881.
- He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2016.
- Khosla, A., Jayadevaprakash, N., Yao, B., and Fei-Fei, L. Novel dataset for fine-grained image categorization: Stanford dogs. In *First Workshop on Fine-Grained Visual Categorization, CVPR*, 2011.
- Russakovsky, O., Deng, J., Su, H., Krause, J., Satheesh, S., Ma, S., Huang, Z., Karpathy, A., Khosla, A., Bernstein, M., Berg, A. C., and Fei-Fei, L. ImageNet large scale visual recognition challenge. *International Journal of Computer Vision*, 115(3):211–252, 2015. doi: 10.1007/s11263-015-0816-y.
- Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 618–626, 2017.
- Wang, H., Wang, Z., Du, M., Yang, F., Zhang, Z., Ding, S., Mardziel, P., and Hu, X. Score-CAM: Score-weighted visual explanations for convolutional neural networks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pp. 111–119, 2020. doi: 10.1109/CVPRW50498.2020.00020.
- Wang, K., Oramas, J., and Tuytelaars, T. MinMaxCAM: Improving object coverage for CAM-based weakly supervised object localization. In *Proceedings of the British Machine Vision Conference (BMVC)*, 2021.
- Zhang, C.-L., Cao, Y.-H., and Wu, J. Rethinking the route towards weakly supervised object localization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 13460–13469, 2020.
- Zhou, B., Khosla, A., Lapedriza, A., Oliva, A., and Torralba, A. Learning deep features for discriminative localization. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 2921–2929, 2016.